Forces and Motion Lab Brief

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Net Force Changes Motion

A force is a push or pull. When the forces on an object are balanced, the net force is zero and the object’s velocity does not change. An object can remain at rest or continue moving at a constant velocity when its net force is zero.

When forces are unbalanced, the net force is not zero. A nonzero net force causes acceleration, which means a change in velocity. The change may be an increase in speed, a decrease in speed, or a change in direction.

Newton’s second law connects net force, mass, and acceleration: $F_{net} = m \times a$. For an object with the same mass, a greater net force produces a greater acceleration. For the same net force, an object with greater mass accelerates less.

Example: A 2 kg cart experiences a net force of 6 N. Its acceleration is 3 m/s² because 6 N ÷ 2 kg = 3 m/s².

Forces in Contact

Friction is a contact force that opposes motion between surfaces. Rougher surfaces usually create more friction than smoother surfaces. Air resistance, also called drag, opposes motion through air. Both friction and drag can reduce an object’s speed when no other force maintains its motion.

The normal force is the support force exerted by a surface. On a level table, the normal force on a resting book acts upward while the book’s weight acts downward. If these forces are equal in size, the net force is zero.

Mass is the amount of matter in an object. Weight is the force of gravity acting on that mass. On Earth, an object with more mass has more weight, but mass does not change when the object moves to a different location.

Example: A hockey puck slows after a player stops pushing it because friction between the puck and the ice creates a force opposite the puck’s motion.

Cart Investigation

Question: How does changing the pulling force affect the acceleration of the same cart?

Trial	Cart mass	Net pulling force	Measured acceleration
1	2 kg	2 N	1 m/s²
2	2 kg	4 N	2 m/s²
3	2 kg	6 N	3 m/s²

Conclusion: With mass held constant, doubling the net pulling force doubled the measured acceleration. The data support the relationship described by Newton’s second law.

Reflection prompts: Which variable was controlled? What evidence supports the conclusion? Predict the acceleration if the 2 kg cart experienced an 8 N net force.

A strong prediction is 4 m/s² because the same 2 kg mass would accelerate in proportion to the net force.